Tech Stack

What language are you writing in?
The language that I will be writing my software in will be Java because my main task is essentially parsing a file, running a data modeling algorithm on it and returning an output file with possibly minimal gui or terminal interface for the initial proof of concept.

What libraries, packages, or external tools do you need? If you're writing Bash, which commands or utilities will you rely on **curl**, **grep**, **jq**, **cron**, **etc.**)? If you're writing Java, which external libraries (if any) do you need?

For my MVP I won't be needing any external libraries but I will be relying on some built in Java tools such as getting input from the user in the terminal, reading the CSV line by line, writing the output.txt, and storing and summarizing data. The java.io class can provide many of the readers, writers, and exceptions that I may need to handle. Java.util provides things like ArrayLists, scanners amongst others. I might also end up using java.nio since it's the modern adaptation of BufferedReaders and is easier to use than a BufferedReader is. In terms of tools outside of code, I will be compiling and running my code with JDK 25 within IntelliJ IDEA as my IDE. For version control I am using Git with its cloud storage in Github. Since some values in the columns might contain a comma and others are done, I will be using a CSV parsing library (Apache Commons CSV). I am not sure right now if I will be using a testing framework, but if I do end up using one then it will be JUnit to make sure my parsing and outputs work well and I will test the fade logic manually using sample CSVs.

How will you store data? For an MVP, a plain text / JSON / CSV file is often enough, so there's no need to jump to a full database.

I will be storing data within a CSV file.

If you're pulling data from an external source, what is it and how will you access it? (See the APIs & External Data Sources post on Ed.)

To get the data within my input CSV I will use a hybrid approach with Exportify and Tunebat. Exportify gives me all the necessary information besides BPM and key. For that specifically, I will be using Tunebat and adding it to the CSV manually before inputting. Both of these outside services are free to use.

Build Plan

- Create a Github Repository for the project, initialize a local Git Repository, set its remote URL to the Github Repository, and set up the environment with the proper compiler (JDK 25) and IDE (IntelliJ IDEA).
- Testing the external services for my input CSV and creating a folder storing a collection of sample input CSVs for later testing and demo.
- Using Apache Commons CSV to parse the CSVs and ensuring its working properly.
- Building the output writer and ensuring the program is able to output a txt file.
- Use JUnit tests to ensure the parser meet program requirements.
- Program the Camelot key logic to give DJ a simple rule-based tip on fading.
- After this, build the algorithm that uses the BPM, key, and energy to give each song pair a transition score. This should also be maximizing the transition score sequence for ordering the output txt file.
- Creating a program that summarizes the data within the output CSV so DJs can easily understand what the output means.
- Use JUnit tests to ensure the summary txt file meets program requirements and handtest the Camelot key algorithm.
- Clean Readme with instructions on how to properly run the program.

System Diagram

